

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
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Section / Batch:	AI&DS	Date:	07.08.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I Mohan Kumar J (192424060) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Mohan Kumar J

Question-2:

Sorting Techniques - Comparative Operations

Problem Statement:

Analyse performance of Selection & Bubble Sort
[57, 21, 43, 12, 36, 18]

Apply both step-by-step, display intermediate array after each pass, count comparisons, swaps.

i) Problem interpretation:

Dataset is unsorted

The objective is to sort it in ascending order using two different comparison-based sorting algorithms & compare their operational performance.

ii) Working Principle:

a) Selection Sort:

Selection Sort repeatedly selects the smallest element from the unsorted portion & places it at correct position.

Characteristic:

→ Fewer swaps

→ Fixed number of comparisons

→ Not stable

→ In-place sorting

b) Bubble Sort:

Bubble Sort repeatedly compares adjacent elements and swaps them whenever they are in wrong order.

[57, 21, 43, 12, 36, 18]
iii) Selection Sort

Pass-1:

min = 12

[12, 21, 43, 57, 36, 18]

Pass-2:

min = 18

[12, 18, 43, 57, 36, 21]

Pass-3

min = 21

[12, 18, 21, 57, 36, 43]

Pass-4

min = 36

[12, 18, 21, 36, 57, 43]

Pass-5

min = 43

[12, 18, 21, 36, 43, 57]

iv) Comparison & swap:

Number of comparisons: 15

Number of swaps: 5

Sorted Array = [12, 18, 21, 36, 43, 57]

v) Bubble Sort:

12 18 21 36 43 57

i j

57 21 43

12 18 21 36

if remainder same
next - 1

21 57 36
 21 43 57 12 36 18
 21 43 12 57 36 18
 21 43 12 36 57 18
 21 43 12 36 18 57

Pass-2:

21 43 12 36 18 57
 21 43 12 36 18 57
 21 12 43 36 18 57
 21 12 36 43 18 57
 21 12 36 18 43 57

Pass-3:

21 12 36 18 43 57
 12 21 36 18 43 57
 12 21 18 36 43 57

Pass-4:

12 21 18 36 43 57
 12 21 18 36 43 57
 12 18 21 36 43 57

∴ Final sorted array = [12, 18, 21, 36, 43, 57]

Total No. of comparisons = 15

Total No. of swaps = 11

vi) Performance Comparison and Time complexity

Parameter	Selection Sort	Bubble Sort
Comparisons	15	15
Swaps	5	10
Best Case	$O(n^2)$	$O(n)$
Average Case	$O(n^2)$	$O(n^2)$
Worst Case	$O(n^2)$	$O(n^2)$
Stability	No	Yes
Memory	$O(1)$	$O(1)$

vii) Interpretation & Justify:

Both successfully sorted into ascending order.

Both required 15 comparison

Selection sort performed 5 swaps

Bubble Sort required 10 swaps.

Since Selection Sort is operationally more efficient for this Dataset. Bubble Sort is advantageous only when stability or nearly sorted data